

Plant Propagation

Science

May, 2026

Plant Propagation (A09835)

Short Title: Plant Propagation
Faculty Science and Computing
Department Science
Cluster Horticulture
Author PFITTERS
Credits 5

Level: Introductory

Description of Module / Aims

A module to enable the student to acquire the principles, knowledge and skills to propagate a range of plants from seed, cuttings, grafting, division and layering. The student will be introduced to a range of techniques useful for the small and large scale production of plants.

Programmes

HORT -0004	BSc in Horticulture (WD_SHORT_D)	1 / 2 / M
HORT -0004	BSc in Horticulture (WD_SHORT_D)	2 / 4 / E
HORT -0004	BSc in Horticulture (WD_SHORT_DX)	1 / 2 / M
HORT -0004	BSc in Horticulture (WD_SHORT_DX)	2 / 4 / E

Indicative Content

- Principles of propagation from seed: Preparing seed for sowing, sowing media, sowing seed, and raising plants
- Types of cuttings, systems, requirements for rooting, and cutting skills
- Principles of grafting and budding: Techniques, equipment required, and grafting skills
- Principles of division and layering: Division operations, and layering operations
- Tissue culture principles and techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use a range of propagation techniques such as growing plants from seed, cuttings, grafting or budding, division and layering.
2. Describe the production of plants from seed, cuttings, grafting or budding, division and layering.
3. Describe the procedure of plant production by tissue culture.
4. Operate safely in a nursery work environment.

Learning and Teaching Methods

- Lectures
- Practical demonstrations
- Site visit

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	2,3
Continuous Assessment	50%	
<i>Practical</i>	50%	1,4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical tasks.

40%-49%: Demonstrates a limited knowledge of the subject matter. Shows some technical ability/skill in carrying out and reporting on practical tasks.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Logically and competently carries out practical tasks.

60%-69%: Demonstrates sound knowledge of the subject matter. Practicals and reports completed to high standard.

70%-100%: Demonstrates excellent technical ability, awareness, and aptitude of the propagation of plants. Delivers insightful reports. Develops high competency in practical areas.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- McDonald, B. *Practical woody plant propagation for nursery growers*. USA: Timber Press, 2006.

Supplementary Material

- MacMillan, B.P. *Plant Propagation*. UK: RHS, 2002.